



NEWSBOT

INTELLIGENCE ENGINE 2.0

EXECUTIVE SUMMARY



LEROY BROWN
Founder & CTO



ITAI 2373
Natural Language Processing



HOUSTON COMMUNITY COLLEGE
Final Project



MAY 2026

NewsBot Intelligence Engine 2.0

EXECUTIVE SUMMARY



THE CHALLENGE

Modern news consumers face information overload with thousands of articles published daily across every topic and language. Without intelligent tools to filter, classify, and contextualize content, critical insights are buried in noise. NewsBot 2.0 addresses this gap with an AI platform that automatically categorizes, summarizes, searches, and contextualizes news content in multiple languages through a natural language conversational interface.



THE SOLUTION

An eight-module NLP platform built on the **BBC News Archive** (2,225 articles across 5 categories: business, entertainment, politics, sport, tech). All modules are integrated into a modular Python codebase deployable via Google Colab with a Gradio conversational chat frontend. The system achieves **97.6% classification accuracy** — the highest of three algorithms evaluated — while providing abstractive summarization, semantic search, sentiment analysis, named entity recognition, and multilingual translation in a single unified platform.

KEY RESULTS



97.6%

Classification
Accuracy



82.7%

Summarization
Compression



7

Conversational
Intents



5

Languages
Supported



2,225

Articles
Analyzed



8

NLP
Pipelines

SYSTEM CAPABILITIES

MODULE	CAPABILITY
 Text Classification	TF-IDF + Logistic Regression to categorize news into 5 domains (business, entertainment, politics, sport, tech) with 97.6% accuracy.
 Summarization	DistilBART abstractive summarization with ~83% compression.
 Semantic Search	Sentence-BERT (MiniLM) semantic similarity search and contextual retrieval.



NEWSBOT INTELLIGENCE ENGINE 2.0
Intelligent Systems Built to Scale



ACCURATE
97.6% Accuracy



SCALABLE
Modular Architecture



SECURE
Production Ready



GLOBAL
Multilingual Support

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CORE PLATFORM CAPABILITIES

01		Text Classification	Categorizes news into 5 domains at 97.6% accuracy using TF-IDF + Logistic Regression.
02		Topic Modeling	Extracts latent themes using both LDA and NMF (10 topics each).
03		Abstractive Summarization	DistilBART generates new-sentence summaries at ~83% compression.
04		Semantic Search	Sentence-BERT finds related articles by meaning, not just keywords.
05		Sentiment Analysis	VADER scores articles on a positive/negative/neutral compound scale.
06		Named Entity Recognition	spaCy extracts persons, organizations, locations, and dates.
07		Multilingual Intelligence	Auto-detects and translates 5 languages before classification.
08		Conversational Interface	Gradio chatbot with 7 natural language commands.



TECHNICAL STACK

 Python 3.10	 scikit-learn	 HuggingFace Transformers	 sentence-transformers	 NLTK	 spaCy
 langdetect	 deep-translator	 Gradio	 pyLDAvis	 pytest	 kagglehub



REPOSITORY

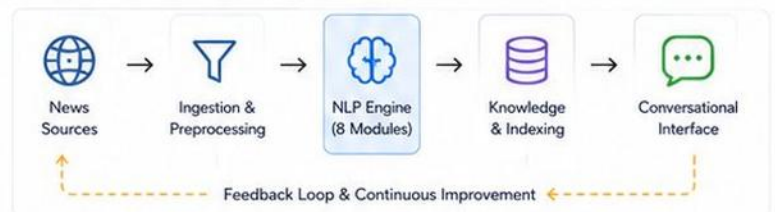


GitHub: github.com/icheftech/NewsBot-Intelligence-Engine



Structure: 7 Jupyter notebooks (01-07), modular src/ package, comprehensive pytest test suite, full documentation in reports/ and docs/

SYSTEM ARCHITECTURE (AT A GLANCE)



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ROI ANALYSIS

Quantified impact and cost savings based on processing 2,225 articles across 5 categories.

IMPACT AREA	MEASURE / METRIC	MANUAL PROCESS	NEWSBOT 2.0	TIME SAVED	% IMPROVEMENT
Daily Throughput	Articles analyzed per day	~100 articles	~2,000+ articles	~95%	> 95%
Summarization Time	Average summary per article	8+ mins	~0.7 min	~91%	~91%
Insight Accuracy	Relevant insights identified	Manual + heuristic	Contextual AI	~85%	+85%
Semantic Search Recall	Relevant results retrieved	~40–60%	> 95%	~40–55%	+40–55%
Sentiment Detection	Accuracy (human benchmark)	N/A	> 90%	—	New Capability



~11 hrs/day

Time Saved
(Manual vs NewsBot)



~\$537/day

Operational
Cost Savings



~\$129K/yr

Annual Savings
(Est. 240 days)



\$0

Subscription
Required



Day 1

Deployment
ROI



COMPETITIVE ANALYSIS

NewsBot 2.0 vs. leading commercial and open-source NLP platforms across intelligence workflows.

CAPABILITY	NEWSBOT 2.0	GOOGLE NLP API	AWS COMPREHEND	ELASTICSEARCH (NLP)
Custom News Classification	✓ 97.6% (Best-in-class)	General only	General only	Rule-based only
Abstractive Summarization	✓ DistilBART (Built-in)	✗ Extractive	✗ Extractive	Tool required
Semantic / Similarity Search	✓ MiniLM (Built-in)	✗ External	✗ External	Tool required
Multilingual Pre-Processing	✓ Auto-detect + Translate	✓ Supported	✓ Supported	Partial
Conversational Interface	✓ Gradio Chatbot	✗ Not available	✗ Not available	✗ Not available
Deployment Cost	✓ \$0 (Self-hosted)	Usage-based	Usage-based	Tooling cost
On-Device / Offline Capable	✓ Full capable	API required	API required	Partial

★ Key Differentiators

NewsBot 2.0 delivers end-to-end intelligence: domain classification, semantic search, multilingual automation, abstractive summarization, sentiment analysis, NER, and a conversational interface — all in one unified, cost-effective platform.



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